Seongju Lee

Ph.D. in AI Convergence

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Google Scholar



SUMMARY

Senior Researcher at HD Hyundai Robotics working on robot learning and simulation for humanoid and dual-arm robots. I build teleoperation systems, multimodal demonstration pipelines, and environments for testing learned policies. My background spans deep learning model design and robust visual perception, with experience integrating learned models into autonomous systems.

SKILLS

Robot Learning: Imitation Learning, VLA, WAM
Simulation: Isaac Sim, MuJoCo
Perception: Detection, Segmentation, Tracking, 6D Object Pose Estimation
Learning: Supervised Contrastive / Metric Learning, Multimodal Learning, Domain Generalization, Knowledge Distillation
Programming: Python, C++, C, MATLAB
Frameworks: PyTorch, ROS/ROS 2, LeRobot
Foundation Models: VFM, LLM, VLM
Robot Platforms: UR5, Franka Panda, HD Hyundai Robotics dual-arm robot
Languages: Korean (Native), English

RESEARCH EXPERIENCE

Jul. 2026 – Present
Senior Researcher — **HD Hyundai Robotics**
Developing robot learning pipelines and simulation environments for intelligent dual-arm robots.

Mar. 2026 – Jun. 2026
Postdoctoral Researcher — **GIST Institute of Artificial Intelligence**

EDUCATION

Mar. 2020 – Feb. 2026
Gwangju Institute of Science and Technology (GIST) — **Advisor: Prof. Kyoobin Lee**
Ph.D. in AI Convergence
Dissertation Title: Robust Perception and Trajectory Forecasting for Autonomous Navigation in the Wild

Mar. 2018 – Feb. 2020
Gwangju Institute of Science and Technology (GIST) — **Advisor: Prof. Kyoobin Lee**
M.S., School of Integrated Technology (Intelligent Robotics Program)

Mar. 2014 – Feb. 2018
Gwangju Institute of Science and Technology (GIST) — **Advisor: Prof. Kwanheng Lee**
B.S. in Mechanical Engineering
UC Berkeley Summer Session (Jul. 2015 – Aug. 2015): Linear Algebra and Differential Equations, Social Inequalities

Mar. 2012 – Feb. 2014
Gangwon Science High School
Early Graduation

PROJECTS

Jul. 2026 – Present
AI & Software Developer
Development of an Intelligent Dual-Arm Robot
HD Hyundai Robotics
TL;DR: Robot learning with VLA/WAM models; VR/glove teleoperation and hand retargeting; simulation and multimodal demonstration data pipelines. [News]

May 2025 – Feb. 2026
AI Developer
Development of Tactile Intelligence in Robotic Hands Based on Tactile Data Learning for Manipulating Irregular Multiple Types of Objects
Ministry of Trade, Industry and Energy (MOTIE), South Korea
TL;DR: Multimodal Action Policy for Contact-Rich Tasks using Vision and F/T Sensing

Sep. 2023 – Feb. 2026
Project Member
Software Developer
Development of a Micro Intelligent Drone for Autonomous Missions in Communication/GPS-denied Environments
Institute of Civil-Military Technology Cooperation (ICMTC), South Korea
TL;DR: RGB-IR object detection and a multimodal RGB, IR, and LiDAR database captured from drone views.

Projects (continued)

Apr. 2023 – Dec. 2024
Project Member
Software Dev. Leader

Development of a Situational Awareness System for Preventing Collisions and Accidents of Autonomous Ships
Ministry of Trade, Industry and Energy (MOTIE), South Korea
TL;DR: Multi-camera maritime object detection and tracking, training dataset creation, and integration into autonomous ship systems.

Apr. 2020 – Nov. 2024
Project Leader
AI & Software Dev. Leader

Development of Swarm USV Autonomous Navigation Algorithm
Agency for Defense Development (ADD), South Korea
TL;DR: Maritime object detection and intention prediction for swarm situational awareness, integrated into multiple USVs. [News]

Mar. 2022 – Dec. 2022
Project Member
AI & Software Dev. Leader

Ceiling Layout Estimation for Indoor Environments
LG, South Korea
TL;DR: Ceiling segmentation and dataset construction for robust floor plan generation under occlusion.

Apr. 2019 – Mar. 2020
Project Member
AI & Software Developer

Robotic IKEA Furniture Assembly from Assembly Manuals
Ministry of Science and ICT (MSIT), South Korea
TL;DR: Assembly-part instance segmentation and workflow visualization for robot assembly from IKEA manuals. [Video]

Mar. 2018 – Dec. 2019
Project Member
AI Developer

Artificial Intelligence-Based Algorithm Development for Sleep Analysis and Diagnosis of Sleep Apnea Syndromes
Gwangju Institute of Science and Technology (GIST), South Korea
TL;DR: Construction of a Polysomnography Database, Development of a Sleep Stage Classification Model, and Development of a Sleep Apnea Detection Model

Mar. 2019 – Aug. 2019
Project Member
AI & Software Developer

Deep Learning-Based Sound Event Detection in Game Environments for Haptic Feedback Generation
Dongwoon Anatech, South Korea
TL;DR: Construction of a Sound Event Detection Dataset and Development of a Sound Event Detection Model in Game Environments

PUBLICATIONS

International Conference and Workshop Papers (*: Equal Contribution)

Top Conference

ManipForce: Force-Guided Policy Learning with Frequency-Aware Representation for Contact-Rich Manipulation
2026 IEEE International Conference on Robotics and Automation (ICRA 2026)
TL;DR: Imitation Learning, Multimodal Policy, Vision Foundation Models (VFMs), Force/Torque Sensor
Geonhyup Lee, Yeongjin Lee, Kangmin Kim, **Seongju Lee**, Sangjun Noh, Seunghyeok Back, Kyoobin Lee
[Paper] [Website] [Code] 38.04% Acceptance Rate Cited by 29

Top Conference

CD-FKD: Cross-Domain Feature Knowledge Distillation for Robust Single-Domain Generalization in Object Detection
2026 IEEE International Conference on Robotics and Automation (ICRA 2026)
TL;DR: Knowledge Distillation, Robust Object Detection, Single-Domain Generalization
Junseok Lee*, Sungho Shin*, **Seongju Lee**, Kyoobin Lee
[Paper] [Code] 38.04% Acceptance Rate Cited by 1

Workshop

4th Workshop on Maritime Computer Vision (MaCVi): Challenge Overview
2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2026), pp. 8146–8156
Benjamin Kiefer et al. (including **Seongju Lee**)
[Paper] Cited by 2

Conference

MV2: A Large-Scale 360-degree Multi-View Maritime Vision Dataset for Object Detection and Segmentation
The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)
Junseok Lee*, Jongwon Kim*, **Seongju Lee*** (Co-first author), Taeri Kim, and Kyoobin Lee
[Paper] [Website]

Conference

Robust Maritime Object Detection under Adverse Conditions via Joint Semantic Learning without Extra Computational Overhead
The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)
Junseok Lee, **Seongju Lee**, Jongwon Kim, Jumi Park, and Kyoobin Lee
[Paper] 46% Acceptance Rate Cited by 2

Conference and Workshop Papers (continued)

Workshop

3rd Workshop on Maritime Computer Vision (MaCVi) 2025: Challenge Results
2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW 2025), pp. 1542–1569
Benjamin Kiefer et al. (including **Seongju Lee**)
[Paper] Cited by 11

Top Conference

MART: MultiscAle Relational Transformer Networks for Multi-agent Trajectory Prediction
The 18th European Conference on Computer Vision (ECCV 2024), Poster
TL;DR: Graph Neural Networks (GNNs), Hypergraph Transformer
Seongju Lee (First author), Junseok Lee, Yeonguk Yu, Taeri Kim, and Kyoobin Lee
[Paper] [Code] 27.9% Acceptance Rate Cited by 80

Conference

Buoy Light Detection and Pattern Classification for Unmanned Surface Vehicle Navigation
The 21st International Conference on Ubiquitous Robots (UR 2024)
Junseok Lee, Taeri Kim, **Seongju Lee**, Jumi Park, and Kyoobin Lee
Cited by 2

Top Conference

Block Selection Method for Using Feature Norm in Out-of-Distribution Detection
2023 IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR 2023), Poster
Yeonguk Yu, Sungho Shin, **Seongju Lee**, Changhyun Jun, and Kyoobin Lee
[Paper] [Code] 25.8% Acceptance Rate Cited by 82

Conference

GYM-Net: Graphical Y-branched Multi-tasking Network for Trajectory Prediction
The 22nd International Conference on Control, Automation and Systems (ICCAS 2022)
Taeri Kim, Junseok Lee, **Seongju Lee**, and Kyoobin Lee

Conference

Vessel Trajectory Forecasting on Naval Warfare Simulation via CVAE-based Model
The 9th International Conference on Robot Intelligence Technology and Applications (RiTA 2021)
Seongju Lee* (Co-first author), Yeonguk Yu*, Junseok Lee, Jumi Park, Changhyun Jun, and Kyoobin Lee

Conference

Real-time Vibro-Acoustical Feedback for the Reality in Game by Detecting the Gun-Sound using Simple Convolutional Neural Network
The 8th International Conference on Smart Media & Application (SMA 2019)
Sungho Shin, **Seongju Lee**, Changhyun Jun, and Kyoobin Lee

International Journal Articles (*: Equal Contribution)

Journal metrics: 2025 JIF and best JCR subject-category quartile.

Journal

Human-efficient annotation system via active semi-supervised learning and selective review
Intelligent Systems with Applications, Vol. 31, Article 200701, 2026
Jongwon Kim, Youmin Park, Seunghyeok Hyun, Yunjae Heo, **Seongju Lee**, Yeonguk Yu, and Kyoobin Lee
[Paper] JCR 2025 Q1

Journal

Automated diagnosis for extraction difficulty of maxillary and mandibular third molars and post-extraction complications using deep learning
Scientific Reports, 2025
Junseok Lee, Jumi Park, **Seongju Lee**, Seong-Yong Moon, and Kyoobin Lee
[Paper] JIF 4.9 (2025, Q1) Cited by 6

Journal

SleePyCo: Automatic Sleep Scoring with Feature Pyramid and Contrastive Learning
Expert Systems with Applications, 2024
Seongju Lee (First author), Yeonguk Yu, Seunghyeok Back, Hogeon Seo, and Kyoobin Lee
[Paper] [Code] JIF 9.4 (2025, Q1) Cited by 107

Journal

BattleSound: A Game Sound Benchmark for the Sound-Specific Feedback Generation in a Battle Game
Sensors, 2023
Sungho Shin, **Seongju Lee**, Changhyun Jun, and Kyoobin Lee
[Paper] [Website] JIF 4.0 (2025, Q2) Cited by 1

Journal

Robust Skin Disease Classification by Distilling Deep Neural Network Ensemble for the Mobile Diagnosis of Herpes Zoster
IEEE Access, 2021
Seunghyeok Back*, **Seongju Lee*** (Co-first author), Sungho Shin, Yeonguk Yu, Taekyeong Yuk, Saepomi Jong, Seungjun Ryu, Kyoobin Lee
[Paper] JIF 4.2 (2025, Q2) Cited by 80

International Journal Articles (continued)

Journal
Self-Supervised Transfer Learning from Natural Images for Sound Classification
Applied Sciences, 2021
Sungho Shin, Jongwon Kim, Yeonguk Yu, **Seongju Lee**, Kyoobin Lee
[Paper] JIF 2.9 (2025, Q2) Cited by 28

Journal
Pilot study of a single-channel EEG seizure detection algorithm using machine learning
Child's Nervous System, 2021
Seungjun Ryu, Seunghyeok Back, **Seongju Lee**, Hyeon Seo, Chanki Park, Kyoobin Lee, Dong-Seok Kim
[Paper] JIF 1.3 (2025, Q3) Cited by 24

Journal
Intra- and inter-epoch temporal context network (IITNet) using sub-epoch features for automatic sleep scoring on raw single-channel EEG
Biomedical Signal Processing and Control, 2020
Hogeon Seo*, Seunghyeok Back*, **Seongju Lee*** (Co-first author), Deokhwan Park, Tae Kim, Kyoobin Lee
[Paper] [Code] JIF 5.7 (2025, Q2) Cited by 313

Domestic Publications

Conference
다중 객체의 경로 정보를 이용한 GNN 기반의 객체 분류 알고리즘 개발
제 19회 한국로봇종합학술대회, KRoC 2024
이상범, **이성주**, 이준석, 김태리, 이규빈

Conference
단일 물체 영역 탐지를 위한 지식 증류 기반의 약지도 학습 프레임워크 개발
2023 ICROS-KROS 호남제주 그랜드 종합 학술대회
김종원, 노상준, **이성주**, 이규빈

Conference
확장된 길이의 다중 채널 신호를 이용한 수면 단계 분류 모델 지식 증류 기법
2023 ICROS-KROS 호남제주 그랜드 종합 학술대회
최소원, **이성주**, 유연국, 이규빈

Conference
Self-attention을 이용한 치과 파노라마 이미지에서의 치아 인스턴스 분할 성능 향상
제 17회 한국로봇종합학술대회, KRoC 2022
박주미, 이준석, **이성주**, 유연국, 이규빈

Journal
물품조립 AI - 조립설명서 기반 로봇의 작업계획 생성 시스템
로봇과 인간 제 18권 (2021)
백승혁, 신성호, 강래영, **이성주**, 이주순, 육태경, 전창현, 김종원, 유연국, 박성호, 윤준호, 이준석, 이규빈

Conference
커리큘럼 학습을 통한 원시 단일 채널 EEG의 자동 수면 단계 분류 성능 향상
제 36회 제어로봇시스템학회, ICROS 2021
박덕환, **이성주**, 백승혁, 이규빈

Conference
표면 법선 벡터 추정 정보를 이용한 2D 와이어프레임 도면 내 개체분할 성능 향상
제 16회 한국로봇종합학술대회, KRoC 2021
이성주 (First author), 전창현, 강래영, 이규빈

Conference
조립설명서에서 구체적인 조립 설명 생성을 위한 문서 분할 데이터셋 구축
제 15회 한국로봇종합학술대회, KRoC 2020
이주순, 육태경, 신성호, 백승혁, **이성주**, 강래영, 최승준, 이규빈

Conference
UWB Radar를 이용한 지능형 호흡 추정 및 무호흡 판별
제 15회 한국로봇종합학술대회, KRoC 2020
박덕환, 서호건, **이성주**, 이규빈

Conference
CRNN 구조를 활용한 뇌전도 신호 기반 수면 단계 분류
제 14회 한국로봇종합학술대회, KRoC 2019
이성주 (First author), 백승혁, 서호건, 이규빈

Technical Reports and Preprints

Technical Report
Object Detection for Understanding Assembly Instruction Using Context-aware Data Augmentation and Cascade Mask R-CNN
arXiv:2101.02509, 2021
Joosoon Lee, **Seongju Lee**, Seunghyeok Back, Sungho Shin, and Kyoobin Lee
[Paper] Cited by 10

Manuscripts and Work in Progress

In Submission

Rethinking Deformable Convolution as an Adapter with Cross-layer Weight Sharing for Robust Semantic Segmentation in the Wild

Target Journal: IEEE Robotics and Automation Letters

First Prize in the USV-based Obstacle Segmentation Challenge at the 3rd Maritime Computer Vision Workshop

TL;DR: Vision Foundation Models (VFMs), Domain Generalized Semantic Segmentation (DGSS), Deformable Convolution Networks (DCNs), Parameter-Efficient Fine-Tuning (PEFT)

Seongju Lee (First author), Raeyoung Kang, Sangbeom Lee, Sangjun Noh, Yeonguk Yu, Junseok Lee, and Kyoobin Lee

WIP

Sensing and Acting Like Human: Learning Contact-Rich Precision Skills from Human-Guided Demonstrations

TL;DR: Multimodal Bimanual Action Policy Model, Cross-Modal Alignment of Vision and Force–Torque

AWARDS

Jun. 2026 — **Multimodal Semantic Segmentation Challenge, 4th Workshop on Maritime Computer Vision (MaCVi)** — IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)
First Place

Jan. 2026 — **RoCo Challenge @ AAAI 2026: Robotic Collaborative Assembling for Human-Centered Manufacturing** — AAAI, A*STAR, NTU
Finalist Excellence Award; integrated 6D pose estimation (MegaPose + ICP).

Nov. 2025 — **Autonomous Driving AI Challenge (Semantic Segmentation)** — MSIT, IITP, KADIF
Second Prize (IITP President's Award)

Dec. 2024 — **USV-based Obstacle Segmentation Challenge, 3rd Maritime Computer Vision Workshop** — IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025)
First Prize

Dec. 2024 — **USV-based Panoptic Segmentation Challenge, 3rd Maritime Computer Vision Workshop** — IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025)
Second Prize

Sep. 2022 — **The first NIA-GIST AI-HUB AI Competition** — NIA, GIST
First Prize

Mar. 2020 — **Presidential Scholarship (GFP)** — GIST

Nov. 2017 — **Track 1, The first GIST Innovative Convergence Technology Contest** — GIST
President's Award

Nov. 2017 — **Track 2, The first GIST Innovative Convergence Technology Contest** — GIST
Dean of Mechanical Engineering's Award

PATENTS

2024 Applied (KR) — **그래프 트랜스 포머 기반 다중 관계 분석 네트워크를 통한 다중 객체의 경로 예측 방법 및 장치** — 2024-0159740
이성주, 이준석, 유연국, 김태리, 이규빈

2024 Registered (KR), Applied (PCT) — **부품조립계획장치** — 10-2647402, PCT/KR2022/000032
이성주, 백승혁, 강래영, 박성호, 신성호, 육태경, 이주순, 이규빈

2024 Registered (KR), Applied (PCT) — **실환경 물체 인식 방법 및 장치** — 10-2662978, PCT/KR2022/008146
이성주, 신성호, 강래영, 박성호, 이규빈

2021 Registered (KR) — **비 접촉 센서를 이용한 신체 상태 정보 획득 방법** — 10-2249237
이성주, 서호건, 박덕환, 김문상, 이규빈

2019 Applied (KR) — **부품조립도 이해와 조립계획 생성 시스템** — 2019-0153768
이성주, 백승혁, 강래영, 박성호, 신성호, 육태경, 이주순, 이규빈

ACADEMIC EXPERIENCE & SERVICE

Mar. 2024 – Dec. 2025 Leader — **Lab Server Administrator** — GIST
Managed GPU servers and developed a real-time GPU/storage monitoring dashboard.

Mar. 2022 – Feb. 2025 — **Technical Research Personnel** — GIST

Mar. 2019 – Dec. 2021 — **Teaching Assistant** — GIST
Advanced Deep Learning (Fall 2021) and Deep Learning (Spring 2019, 2020, 2021)

Dec. 2015 – Feb. 2018 — **Early Research Internships** — GIST
Artificial Intelligence Laboratory, Healthcare Robotics Laboratory, and Intelligence Design & Graphics Laboratory